



Estuaries and mudflats

Where rivers reach the sea, they often broaden out into tidal estuaries. The salty seawater makes the tiny mud particles carried in the river water settle on the bottom in thick layers, which are exposed as mudflats at low tide. The mud is salty and airless but, despite this, it is rich in food and home to huge numbers of animals.

MUDDY RIVERS

The mud particles carried by rivers are microscopic, but the salt in seawater makes them clump together into bigger, heavier particles that sink to the river bed. When the tide is rising, the incoming flow of seawater stops the river water moving, which also encourages particles to settle.

TIDAL BORES

When the rising tide pushes water up an estuary to where the river is narrower, this can cause a funnel effect that forces the water level to rise higher and higher. On some rivers this creates a wave that surges upriver, called a tidal bore. Some of these are high enough to surf on.



▲ RÍO DE LA PLATA

This view from space shows the muddy water of a South American river forming a broad estuary where it meets the sea.

GLEAMING MUDFLATS

The muddy sediment that settles on the river bed at high tide is exposed as mudflats when the tide goes out. The falling tide allows the river flow to speed up, so it scours a narrow central channel through the gleaming mud. This is joined by many smaller channels flowing off the flats.

► NATURAL PATTERN

Tiny drainage channels carry water off the mud into bigger channels, which join up to flow into the main river.

WOW!

Twenty-two of the world's biggest cities are located on river estuaries, including London, New York, and Shanghai.